# BiteSwift

## 1.Project Title

BiteSwift – Real-Time Restaurant Ordering & Delivery Management System with AI-Powered Smart Recommendations

## 2. Introduction

BiteSwift is a full-stack, white-label, real-time food ordering and delivery system built using the MERN stack, designed exclusively for independent restaurants. The platform empowers restaurants to accept online orders, manage kitchen operations, track deliveries, and retain full ownership of customer data and revenue — without depending on third-party aggregators.

Key highlights include dedicated dashboards for customers, kitchen staff, delivery riders, and admins; real-time order flow via Socket.IO; secure Stripe payments; OTP-based delivery verification; and AI-powered smart menu recommendations using TensorFlow.js. This intelligent feature provides personalized “Recommended for You” suggestions based on customer order history and semantic understanding of menu items, boosting upsell opportunities and enhancing user experience — all while running client-side in pure JavaScript.

## 3. Problem Statement

8. Core Features

1. Menu Management (Admin)
    - Categories, items, add-ons, availability toggles
    - Image uploads via Cloudinary
    - Price & customization controls
2. Customer Ordering Interface
    - Dynamic menu display with categories
    - Cart management, address input, delivery notes
    - Payment options: Stripe or Cash on Delivery
3. Smart Menu Recommendations (AI-Powered)
    - Personalized “Recommended for You” section on menu page
    - Powered by TensorFlow.js Universal Sentence Encoder (pre-trained, runs in browser)
    - Analyzes customer past orders and menu item names/descriptions semantically
    - Suggests similar items (e.g., chicken lovers see other chicken dishes)
    - Falls back to popular items for new/guest users
    - Fully client-side: fast, private, offline-capable, no external AI API costs
4. Real-Time Synchronization (Socket.IO)
    - Instant kitchen notifications,statusupdates, live tracking
5. Delivery Workflow Automation
    - Auto rider assignment, OTPverification

Traditional food delivery platforms charge high commissions, restrict direct customer relationships, and offer limited control over operations. Additionally, most online ordering systems lack intelligent personalization, missing opportunities to increase average order value through relevant suggestions.

Restaurants need a direct, commission-free system that not only streamlines operations but also intelligently recommends menu items to customers based on their preferences — all while maintaining privacy and performance.

BiteSwift solves these with a complete self-hosted ecosystem featuring on-deviceAI recommendations powered by TensorFlow.js.

4. Objectives

- Develop a real-time online ordering platform using the MERN stack
- Enable end-to-end order lifecycle management (ordering → kitchen → delivery)
- Provide role-specific dashboards for admin, kitchen, riders, and customers
- Implement real-time updates using Socket.IO
- Support secure Stripe payments and Cash on Delivery
- Enable automated rider assignment and OTP verification
- Implement AI-powered smart menu recommendations using TensorFlow.js (client-side, no external API)
- Ensure fully responsive, white-label, and privacy-focused customer storefront

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6. Payment System
    - Secure Stripeintegration + COD
7. Admin Dashboard
    - Live orders,sales analytics, user/rider management
8. Additional Features
    - Responsive design, sound/visual alerts, custom domain support

9. Key Technical Highlights

- Real-time architecture with Socket.IO
- Scalable REST APIs with validation
- Secure Stripe webhooks
- Cloudinary for optimized images
- Efficient MongoDB operations
- Role-based JWT authentication
- Client-side AI recommendations using TensorFlow.js (Universal Sentence Encoder)
- Zero external AI APIdependency — privacy-first and cost-effective

10. Expected Outcomes

BiteSwift delivers:

- A complete, real-time, commission-free ordering ecosystem
- Higher average order value through intelligent, personalized recommendations
- Faster and smarterkitchen-to-delivery workflow

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- Full data ownership and revenue retention
- A modern, production-ready MVP with cutting-edge AI integration
- An impressive full-stack + AI portfolio project

## 11. Conclusion

BiteSwift stands out as a powerful, independent alternative to third-party platforms by combining real-time operations, secure payments, automated delivery, and AI-driven smart recommendations — all in a clean, scalable MERN stack application.

With TensorFlow.js enabling on-device intelligence, restaurants gain personalized upselling capabilities without sacrificing privacy or incurring recurring AI costs. This combination of robust full-stack engineering and modern client-side machine learning makes BiteSwift not just functional, but truly smart and future-ready.